# Engineering Mathematics III

ENGE702

Lecture 2: Mathematical modelling and ODEs

# Differential Equations

In the last lecture we checked that a function was a solution to an ODE, but how do we find these solution functions?

It isn't always possible to find nice functions that are solutions to differential equations, even if we know such a solution exists. In general there are three types of solutions we can get:

1. **Analytic**: sometimes, for certain 'nice' differential equations, we can find an exact formula for the solution.
2. **Numerical**: for many differential equations, we can find approximate solutions, which may be good enough for practical applications.
3. **Qualitative**: often we can describe the nature of important features of the solutions.

In this course we will look at all of these kinds of solutions.

# Differential Equations

Whether you know it or not, you already know how to solve some differential equations. Any differential equation of the form

$$y' = f(x)$$

can be solved by integrating $f(x)$ (as long as we actually can integrate $f(x)$).

**Example**: Suppose $y' = 2x^2 + \cos(2x)$. Writing this another way, we have:

$$\begin{aligned} \frac{dy}{dx} &= 2x^2 + \cos(2x) \\ dy &= \left(2x^2 + \cos(2x)\right)\, dx \end{aligned}$$

Now integrate both sides:

$$\begin{aligned} \int dy &= \int 2x^2 + \cos(2x)\, dx \\ y &= \frac{2x^3}{3} + \frac{1}{2}\sin(2x) + c \end{aligned}$$

# Differential Equations

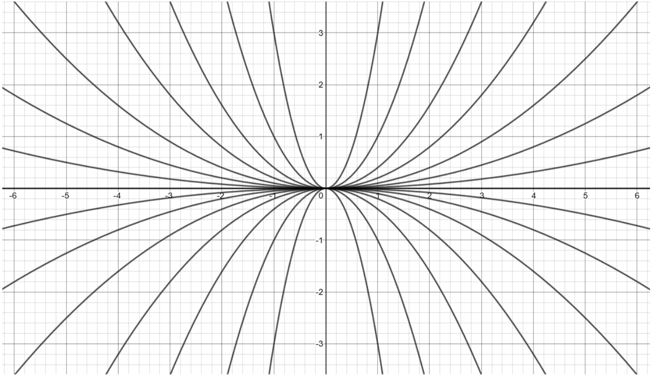
**Exercises**: Solve the following differential equations:

- $y' = x^3 - 2x^2 + 1$
- $y' - 2\sin(x) = 1$
- $y'' = 2x$

# Differential Equations

In a previous slide we saw that $y = cx^2$ is a solution to $xy' = 2y$. Where does the $c$ fit in?

For each real number $c$, there is a solution. A solution of this kind is actually a family of solutions. This is called a **general solution**.



# Differential Equations

In order to find a **particular solution** (turn the $c$ into a number), we need to know some more information. This is usually given in the form of an **initial condition**.

An initial condition specifies the value of the solution at a certain time. Once we have a general solution we can work out $c$ using the initial condition.

A differential equation together with an initial condition gives an **initial value problem (IVP)**.

**Example** Solve $y' = \sin(3x)$, where $y(0) = 1$.

First find the general solution. We can find this by integrating, $y = -\dfrac{1}{3}\cos(3x) + c$.

Now we use the fact that $y(0) = 1$. This means that when we substitute $0$ for $x$ in the general solution, we need to get $1$ as the result.

# Differential Equations

So we have:

$$\begin{aligned} 1 &= -\frac{1}{3}\cos(3 \cdot 0) + c \\ 1 &= -\frac{1}{3} + c \\ c &= \frac{4}{3} \end{aligned}$$

So now we have a value for $c$, so the particular solution to the initial value problem is:

$$y = -\frac{1}{3}\cos(3x) + \frac{4}{3}$$

# Differential Equations

**Exercises**: Solve the following initial value problems:

- $y' = x^2 + 1, \quad$ with $y(0) = 0$
- $y' = e^{3x}, \quad$ with $y(0) = 2$

# Modelling the position of the dropped ball

We shall assume that air resistance is practically negligible and that the only force acting on the ball is $F = mg$ where $g = 9.80$ m/s$^2$ is the acceleration due to gravity towards the centre of the earth and $m$ is the mass of the ball.

Let $y(t)$ be the displacement of the ball dropped from a height, initially at rest. Then we have

$$y'' = g$$

We can work out the solution $y(t)$ by integrating twice and applying our initial conditions, $y(0) = 0$ and $y'(0) = 0$. We won't need the mass $m$. However, if the surface area of the dropped object is very large (such as for a parachute) we can no longer assume that air resistance is negligible, so be careful.

# Modelling the position of the ball

We have

$$y'' = g$$

Velocity $v(t) = y'(t)$ and acceleration $v'(t) = g(t) = g$.
Integrate both sides of the latter equation:

$$v(t) = \int \frac{dv}{dt} dt = \int g \, dt = gt + C_1$$

Since $v(0) = 0$ then $C_1 = 0$ and $v(t) = y'(t) = gt$. Integrate again:

$$y(t) = \int \frac{dy}{dt} dt = \int gt \, dt = \frac{1}{2} gt^2 + C_2$$

Since $y(0) = 0$ then $C_2 = 0$ and $y(t) = \frac{1}{2} gt^2$.

# Parachute problem

**Problem:** Two parachutists leave an aircraft flying horizontally at an altitude of $3000$ m. One has a mass of $85$ kg and one has a mass of $65$ kg. Assume that they leave the aircraft at the same time, under the same windless conditions, and open their parachutes at the same time, far enough away from each other to avoid a collision or interference.

1. How long does it take the parachutists to reach the ground?
2. What is their speed at landing?
3. Who arrives first and why?

We can model this problem in order to find a function of their positions at a particular time. Start by drawing a diagram. We might need some extra information. What parameters would we need to determine? We will develop this together in class at a later date.

# Exercises

1. Verify that $y = c_1 e^{2x} + c_2 e^{-2x}$ is a solution to the differential equation $y'' - 4y = 0$.

2. Solve the following initial value problems:

   - $y' = 3x + 1$, $y(0) = 1$
   - $y' = \frac{1}{x}$, $y(1) = 2$

# Text Book References

Chapter 1.1